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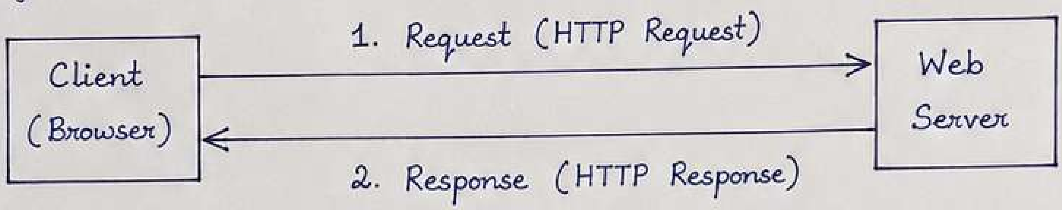
NB 03

Frontend - Assignment 1

Q1. HTTP Request-Response Cycle

Ans: The HTTP Request-Response Cycle is the process in which a client (browser) requests a resource from a web server and the server sends back a response.

Major steps involved are:



1. The client (browser) enters a URL or clicks a link.
2. The browser sends an HTTP request to the web server.
3. The server processes the request.
4. The server sends an HTTP response (HTML, CSS, JS, images, etc.).
5. The browser renders the response and displays the web page.

Q2. Introduction to Web Technologies

Ans:
1. HTML : It is used to create the structure and content of web pages.
2. CSS : It is used to style and layout the web pages (colors, fonts, spacing, etc.).
3. JavaScript : It is used to add interactivity and dynamic behavior to web pages.
4. Bootstrap : It is a CSS framework that helps in building responsive manipulation, event handling, animations, and AJAX.
5. jQuery : It is a JavaScript library for building user interfaces, especially single-page applications, using components.

Q3. Client-Side vs. Server-Side Technologies

Ans:

Client-Side Technologies

- Runs on the client (browser).
- Visible to the user.
- Handles UI and interactions.
- Examples: HTML, CSS, JavaScript, React.js, jQuery.

Server-Side Technologies

- Runs on the server.
- Handles data, logic and database.
- Not visible to the user directly.
- Examples: PHP, Python, Java, Node.js.

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Q4. Block-Level Elements in HTML

Ans: Block-level elements start on a new line and take full width available. They can contain other block or inline elements.

Examples: 1. <div> 2. <p> 3. <h1> to <h6>

Q5. Semantic HTML Elements

Ans: Semantic elements clearly describe their meaning to both the browser and the developer. They improve SEO and accessibility.

Examples: <header>, <nav>, <section>, <article>, <footer>.

Q6. Creating a Table Using HTML

Ans: The table should include appropriate table headings, rows, and columns.

Sr. No.	Name	Designation	Department
1	John	Manager	Sales
2	Smith	Sr. Manager	Marketing
3	Jane	Manager	HR

HTML Code:

```
<table border="1">
    <tr>
        <th> Sr. No. </th>  <th> Name</th>  <th> Designation</th>  <th> Department</th>
    </tr>
    <tr>
        <td> 1 </td>  <td> John </td>  <td> Manager </td>  <td> Sales </td>
    </tr>
    <tr>
        <td> 2 </td>  <td> Smith</td>  <td> Sr. Manager</td>  <td> Marketing </td>
    </tr>
    <tr>
        <td> 3 </td>  <td> Jane </td>  <td> Manager </td>  <td> HR </td>
    </tr>
</table>
```

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Q7. Creating a Nested List Using HTML

Ans: The following is the HTML code for the given nested list.

```
<ul>
  <li> River
     <ul>
        <li> Ganga </li>
        <li> Yamuna </li>
        <li> Brahmaputra </li>
        <li> Narmada </li>
     </ul>
  </li>
</ul>
```

- River
  - Ganga
  - Yamuna
  - Brahmaputra
  - Narmada

Q8. Types of CSS

Ans: There are three types of CSS used to apply styles to an HTML document.

Type	Description	Example
1. Inline CSS	Used inside an HTML element using the style attribute.	This is a paragraph.
2. Internal CSS	Used inside a <style> tag in the <head> section.	<style> p { color: blue; } </style>
3. External CSS	Used in a separate .css file and linked with HTML.	<link rel="stylesheet" href="style.css">

Q9. Practical Understanding

Ans: HTML, CSS and JavaScript are the three main technologies used in web development but each has a different role.

- HTML (HyperText Markup Language) is used to create the structure and content of a web page. It tells the browser what to display.
- CSS (Cascading Style Sheets) is used to style and design the web page. It controls the colors, fonts, layout, spacing, etc.
- JavaScript is used to add interactivity and dynamic behavior to web pages. It can handle events, manipulate elements, validate forms and more.

Together, they create a complete and interactive website.

Q10. Short Answer

Ans:

a. The role of Bootstrap in web development is to provide pre-designed CSS classes and components. It helps in building responsive, mobile-first websites quickly and consistently.

b. jQuery is a JavaScript library that simplifies HTML DOM manipulation, event handling, animations and AJAX. It was widely used because it made JavaScript easier to write and worked across browsers.

c. React.js is a JavaScript library for building user interfaces using components. It solves the problem of managing complex UI and state efficiently, especially in large single-page applications.

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